Test Plan and Results

For evaluating our MVP, we relied on the **Apple Plant Disease Detection model**, trained using the PlantVillage dataset from Kaggle. Since our system is designed to allow users to diagnose plant diseases simply by **showing a leaf to the camera**, we wanted to test it in a realistic, everyday scenario rather than by uploading images manually.

To do this, we opened a **random image of an apple leaf infected with Cedar Apple Rust from Google Images** and showed it to the laptop's webcam. This approach helped us simulate how an actual end-user might use the system at home or in a garden: by pointing their device at a leaf rather than searching and uploading files.

During the test, the model successfully identified the disease as **Cedar Apple Rust**, and did so with a **very high confidence of around 99%**. This gave us strong initial validation that the detection pipeline—camera input, YOLO inference, model processing, and further explanation—was functioning in a stable manner. The bounding boxes were precise, the classification was accurate, and the output text generated by the LLM clearly explained the disease and possible remedies.

But just as importantly, our tests also revealed the **real-world limitations** of the model:

- **The model does not always get predictions right.** In several trials, when the camera was slightly tilted or when the leaf image was too small on the screen, the confidence dropped significantly. This caused the prediction to swing towards other diseases or, in some cases, label the leaf as "healthy" even when mild symptoms were present.
- **Lighting plays a major role.** Under poor lighting or shadows, the model sometimes misread the color patches on the leaf and assigned incorrect labels. Bright reflections also caused the model to miss the disease completely.
- **Generalization is limited.** Since the MVP is trained on a well-curated dataset with clear, centered images, it struggles when symptoms are faint, partially visible, or in early stages. It also only covers apple diseases at the moment, so it cannot correctly classify ornamental plants or less-documented species.
- **Accuracy fluctuates depending on the image.** While we achieved a near-perfect result in our best test case, other trials saw confidence scores drop to

the 60–70% range, showing that the reliability depends heavily on how close the test conditions are to the dataset it was trained on.

These mixed outcomes were actually extremely valuable. The 99% accurate prediction demonstrated that the core system works as intended when the input conditions align well with the training data. Meanwhile, the inconsistencies helped us better understand what needs to be improved as we scale to real-world usage: more diverse training data, better handling of lighting variations, and expansion into new plant types—especially ornamental plants, where available datasets are far more limited.

Overall, the testing phase gave us both the confidence that our approach is viable and a clear roadmap of the challenges we need to address as we move toward a production-ready solution.